

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

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Instructions: Read the questions carefully, show all your work, and explain all your answers. Answer any THREE of the following FOUR questions. Indicate carefully if you introduce notation, or make assumptions when answering a question.

Good luck!

1. Consider the following version of the standard asset-pricing model. There are two trees, whose dividends are perfectly negatively correlated. Specifically, the first tree yields a sequence of dividends $\{z_{1,t} = z_t\}_{t=0}^{\infty}$. The second tree yields a sequence of dividends $\{z_{2,t} = 1 - z_t\}_{t=0}^{\infty}$. The process z_t is Markov with the transition function Q , taking on values in the interval $(0, 1)$.

There is a continuum of individuals of mass one. Each individual maximizes the expected discounted utility of consumption. The instantaneous utility is given by $U(c)$, where c denotes consumption, and the function U is strictly increasing, concave, and continuously differentiable. Individuals discount the future with $\beta \in (0, 1)$. At $t = 0$, each person is endowed with one share of each tree. The consumption good is nonstorable.

- a. Formulate the recursive problem for an individual in this economy. Derive the first-order conditions and the Envelope condition for this problem.
- b. Define a recursive competitive equilibrium.
- c. Derive *explicit* expressions for the prices of the two trees. Interpret the expressions intuitively. Show that $p_1^*(z) + p_2^*(z) = a$, where a is a constant. What is a equal to?

- d. Consider a special case where the dividends follow a binary process with persistence, i.e., $z_t \in \{\bar{z}, 1 - \bar{z}\}$, and $\Pr\{z_{t+1} = \bar{z} | z_t = \bar{z}\} = \Pr\{z_{t+1} = 1 - \bar{z} | z_t = 1 - \bar{z}\} = \pi > 1/2$. For concreteness, let $\bar{z} \in (1/2, 1)$. How does an increase in persistence affect the amount by which asset prices change when z_t changes? Explain your result intuitively.

2. Consider the following endogenous-growth model with government policy. There is a measure one of identical households. Preferences over sequences of consumption $\{c_t\}_{t=0}^{\infty}$ are described by the utility function

$$\sum_{t=0}^{\infty} \beta^t \log(c_t)$$

where $\beta \in (0, 1)$. Each household is endowed with an initial capital stock $\bar{k}_0 > 0$ in the first period and with one unit of labor in each period. Capital depreciates fully after one period.

There is a continuum of measure one of firms indexed by $i \in [0, 1]$. They use capital and labor to produce output, which can be consumed or invested. The production function of firm i is

$$y_{it} = \Phi k_{it}^{\theta} (A_t l_{it})^{1-\theta} \quad (1)$$

where $\Phi > 0$ and $\theta \in (0, 1)$ are constants, k_{it} and l_{it} are the capital and labor of firm i at time t , and $A_t > 0$ is labor-augmenting technological progress. Note that Φ , θ , and A_t are common to all firms, and so they are not indexed by i . A_t is endogenously determined by the aggregate capital stock:

$$A_t = \int_0^1 k_{it} di \quad (2)$$

- a. Define a symmetric, competitive equilibrium with a sequence of markets. Derive the equilibrium consumption growth rate.
- b. Write down the planner's problem and derive the first-order conditions for Pareto-efficient consumption growth. Show that for intermediate values of Φ the equilibrium consumption growth rate is negative whereas the Pareto-efficient consumption growth rate is positive.
- c. Suggest a fiscal policy that uses a labor income tax in combination with a subsidy given to households or firms in such a way that the resulting equilibrium consumption growth rate equals the efficient one.
- d. How does your answer in (c) relates to the results from class about optimal taxation in the basic version of the growth model with exogenously given A_t ? Would you still want to use the fiscal policy from (c) in that basic version? Explain your answer!

3. Consider the following version of the overlapping generations model with capital. The novelty in the current framework is that revenues from tax collections are used to finance public infrastructure, a productive input in production.

Time is discrete ($t = 0, 1, 2, \dots$). A cohort of size L is born every period. There is no population growth. Individuals live for two periods, and work in only the first period of their lives, when they are endowed with one unit of productive time. An individual born at date t chooses consumption levels $c_{1,t}$, $c_{2,t+1}$ and savings s_t , to maximize

$$U(c_{1,t}, c_{2,t+1}) = \ln(c_{1,t}) + \beta \ln(c_{2,t+1})$$

Competitive firms have access to a production technology that uses capital and labor services, and publicly-provided infrastructure capital. All firms have access to infrastructure capital that is provided by a government. A firm that uses l units of labor, k units of capital, given z units of public infrastructure, produces

$$y = z^\lambda k^\alpha l^{1-\alpha}$$

units of output, with $\lambda \in (0, 1)$ and $\alpha \in (0, 1)$. Capital depreciates at the rate $\delta_k \in [0, 1]$.

A government in this economy taxes the labor income of young individuals at the time-invariant rate $\tau \in (0, 1)$ to finance aggregate investment in public infrastructure. Infrastructure capital depreciates at the rate $\delta_z \in [0, 1]$. We assume that firms have access to a level of infrastructure capital equal to the *per-worker* level available in the economy.

- a. Formulate the problem of a young individual in period t . Obtain the first order conditions that characterize its solution. Solve for individual consumption levels and savings.
- b. State clearly the government budget constraint in this economy. Define a competitive equilibrium.
- c. Find the steady-state value for the standard capital-labor ratio.
- d. Use your previous results to evaluate critically the following claim: "Higher rates rates of taxation are needed to finance public infrastructure and therefore, increase the long-run level of investment in private capital."

4. This problem asks you to analyze optimal labor-income taxation in a model without capital. Consider an economy populated by a mass one of identical agents with preferences given by

$$\sum_{t=0}^{\infty} \beta^t \left(c_t - \frac{1}{1+\varphi} l_t^{1+\varphi} \right),$$

where c_t is consumption, l_t is labor supply, and $\beta \in (0, 1)$. There is no capital in this economy, and the production technology is linear in labor: $y_t = l_t$. Each period, the government purchases a variable amount $g_t \leq 1$ of consumption goods, and levies a labor-income tax of rate τ_t . Competitive firms pay wages equal to the corresponding marginal products. Consumers and government can borrow/lend at the gross risk-free rate β^{-1} , and start period $t = 0$ with no assets or liabilities. The government chooses $\{\tau_t\}$ to maximize the representative agent's utility, subject to being able to finance its purchases $\{g_t\}$ with taxes $\{\tau_t\}$ on an intertemporal basis.

Everywhere in the problem, assume that the solution to the consumer's problem is interior.

- a. Write down the first-order conditions for the household's problem.
- b. Substitute from the expressions for tax revenues and prices obtained in a. into the government budget constraint. Your answer should be one equation that depend on the sequences $\{g_t, l_t\}$ and the parameters β and φ .
- c. Using the resource constraint, replace c_t into the agent's intertemporal preferences. Your answer should be one expression that depends on the sequences $\{g_t, l_t\}$ and the parameters β and φ .
- d. Using b. and c., consider the welfare consequences of two fiscal policies. In the first policy, the budget is balanced period by period. In the second policy, tax rates are constant. Which one is better? Why?

